Fiscal Year:	FY 2006	**Task Last Updated:**	FY 08/07/2006
PI Name:	Loewenberg, Michael Ph.D.		
Project Title:	Bubble-scale Modeling of Foams		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	michael.loewenberg@yale.edu	**Fax:**	FY 203 432-4387
PI Organization Type:	UNIVERSITY	**Phone:**	203 432-4334
Organization Name:	Yale University		
PI Address 1:	Department of Chemical Engineering, Mason Laboratory		
PI Address 2:			
PI Web Page:	http://www.eng.yale.edu/		
City:	New Haven	**State:**	CT
Zip Code:	06520-8286	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	02/19/2002	**End Date:**	01/01/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2722		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Interest in foams derives from their broad engineering and technological relevance. Currently, there is an active international effort to elucidate the mechanism of drainage in foams. Simplified macroscopic theories have been proposed but further progress is hampered by the lack of fundamental descriptions for the bubble-scale fluid motion and surfactant transport. Presently, simple models are extensively used to describe the permeability of foams and there has been considerable experimental effort to validate these mechanisms. The current research indicates that the mechanism of foam drainage depends on detailed system parameters which are ignored by the simple models; it thus seems that the existing models are inadequate. Detailed numerical simulations will provide a valuable tool for developing an improved model for the permeability of a foam but none are available. The goal of this project is to further a fundamental understanding of foam drainage by exploring the detailed fluid flow and surfactant transport at the microscopic and mesoscopic scales using numerical simulations. Theoretical results will also be derived. 1. Bubble-scale modeling a) Quasi-one-dimensional flow: Analyze the flow and surfactant transport in slender channels. b) Generalized drainage regime: Develop three-dimensional numerical simulations for the flow and surfactant transport in a Plateau border and in ordered periodic structures, such as the Kelvin cell . 2. Mesoscale modeling a) Permeability of foam networks: Develop Kirchoff-type laws for the permeability (conductivity) of random foam networks using the results of our microscale calculations to supply lumped conductivity parameters. The validity of this approach will be tested. b) Nonlinear effects in deformable Plateau border networks: Compute permeabilities in systems with Plateau borders that respond to the mesoscale spatial pressure fluctuations that arise from asymmetries in foam microstructure. c) Transport in foams: Explore Taylor dispersion resulting from the microscale flow profile in channels and nodes, and from mesoscale spatial fluctuations in flow rates. 3. Foam drainage in microgravity Analyze foam drainage (flow and transport) under the influence of imposed temperature and surfactant-concentration gradients. The results of this project will be tested against the available experimental data, and close contact with experimental groups (e.g., Stone, Durian, Weaire) will be sought. The proposed study will further a fundamental understanding of foam microphysics and should make feasible the development of reliable macroscopic theories which properly incorporate the influence of the relevant system parameters. The simulations and models developed through this study will also help to guide experiments on foam drainage, including new experiments to observe capillary--driven foam drainage. Foam drainage a gravity--sensitive phenomenon which depends on the balance between capillary and buoyancy stresses. Fundamental research on foam microphysics, such as the proposed project, will be therefore be useful for the rational design of foam systems in variable-- and low--gravity environments. The proposed project involves the development of numerical simulations and analysis of novel capillary--driven drainage mechanisms which may be convenient for use in microgravity applications. Space flight experiments using a centrifuge may be useful for observing the influence of variable gravity on forced drainage. Microgravity experiments may be particularly useful for observing capillary--driven foam drainage.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foam drainage: a gravity-sensitive phenomenon which depends on the balance between capillary and buoyancy stresses. Fundamental research on foam microphysics, such as the proposed project, will be therefore be useful for the rational design of foam systems in variable- and low-gravity environments.
Task Progress:	We developed a new theory for foam drainage that correctly includes the effects of surfactant. Our results show that Marangoni stresses hinder foam drainage far less that was previously believed. Our theory predicts drainage rates up to 20 times larger than that predicted by the earlier models which assume no-slip boundary conditions on the air-liquid interfaces. Our analysis shows that drainage rates are sensitive to the interfacial rheology associated with the surfactant. We developed the first three-dimensional boundary integral simulations of foam drainage. Free-surface boundary conditions apply on the liquidgas interface. The equilibrium interface shape is obtained from the Surface Evolver program. Flow-induced interface deformation is neglected under the assumption of small-capillary-number conditions, which apply for the experimental studies that have been performed by various groups (Stone and Harvard; Durian at UCLA). The permeability depends on liquid fraction and tends to a finite value in the zero-liquid-fraction limit, which is relevant to the previous experimental studies. The limiting value differs considerably from that found experimentally by Stone and co-workers. The likely explanation is Marangoni stresses resulting from trace quantities of adsorbed surfactants. Free-surface boundary conditions are likely to apply to systems such as polymer foams and metallic foams.
Bibliography Type:	Description: (Last Updated: 10/07/2019)
Articles in Other Journals or Periodicals	Vlahovska PM, Loewenberg M, Blawzdziewicz J. "Deformation of a surfactant-covered drop in a linear flow: the effects of dropphase viscosity." Submitted 2006.[Ed. note Sep 2019: no online publication found] , Jun-2006

Articles in Peer-reviewed Journals	Vlahovska PM, Loewenberg M. "Deformation of a surfactant-covered drop in a linear flow." Physics of Fluids. 2005 Oct;17(10):103103. https://doi.org/10.1063/1.2112727 , Oct-2005